Appendix II. Debt Sustainability Analysis Update

This joint IMF-World Bank Debt Sustainability Analysis (DSA) updates the analysis presented to the Board in March 2015 and indicates that Sierra Leone's risk of debt distress remains moderate, although the vulnerabilities have significantly increased since the last DSA update. Under the baseline scenario, two ratios, the PV of debt to exports and to revenue, have deteriorated significantly, due to the loss of iron ore exports and the lingering impact of Ebola. These ratios, however, remain under their policy dependent thresholds throughout the projection period (2015–35).[1] Under alternative macroeconomic assumptions, where the economy permanently loses the iron ore sector, the debt sustainability worsens further. The economy is now more vulnerable to adverse shocks to exports, FDI and nominal depreciation.

1. **This DSA updates the analysis presented to the Board in March 2015 and indicates that, while Sierra Leone's debt sustainability situation has worsened, the country remains at moderate risk of debt distress**. The downside risk to the macroeconomic outlook identified in the previous DSA, regarding the potential suspension of iron ore production in 2015, has now materialized. Despite all debt indicators falling below their respective policy-dependent indicative thresholds throughout the projection period (2015–35), loss of iron ore exports and related fiscal revenues, has significantly worsened the short-term ratios of the PV of debt to exports and revenue, respectively. As a result, the room to accommodate more borrowing without breaching the respective thresholds has decreased. Nevertheless, Sierra Leone's moderate risk of debt distress is maintained as none of the ratios breach the thresholds under the baseline scenario on a significant and protracted basis.

2. **The WEO projections of persistently low global iron ore prices, cast huge uncertainty on the future of iron ore production in Sierra Leone**. It is plausible iron ore sector may permanently stop operations over the forecast horizon (2015–35) and the alternative scenario (green line) presents the DSA dynamics under that case with corresponding macroeconomic assumptions. The country's GDP prospects deteriorate over the forecast period with the cessation of iron ore production, with the loss of associated exports and fiscal revenues over the medium term relative to the baseline. The ratios of the PV of debt to exports and debt to revenue exceed their respective thresholds in the long-term, while the debt service to revenue ratio crosses the acceptable ceiling in 2024 due to the redemption profile of external debt.

3. **The DSA shows that the medium- to long-term debt outlook is vulnerable to adverse shocks to several macroeconomic variables**. Shocks to exports, foreign direct investment (FDI) inflows, and nominal currency depreciation would lead to the significant breaches of several thresholds in the short to medium run on a protracted basis. Despite all ratios eventually falling below their respective thresholds in the long run, the realization of an adverse shock may lead to the

[1] Sierra Leone's capacity to monitor debt is adequate as per CPIA with a score of 3.5.

immediate problems of debt distress. Therefore, staff continues to reiterate the need for prudent borrowing policies, revenue enhancement, sustained fiscal consolidation efforts, continued implementation of growth-enhancing structural reforms, and promotion of economic diversification. The authorities agreed with staff's assessment and policy recommendations.

4. **The baseline macroeconomic assumptions underlying this DSA update are**:

- Economic growth is expected to collapse from 4.6 percent in 2014 to -21.5 percent in 2015 due to the impact of the Ebola epidemic and suspension of iron ore production. GDP is expected to stay flat in 2016 and grow by an annual average of 12.4 percent during 2017–20, as iron ore production resumes. Over the long term, growth is expected to stabilize close to the historical rates for the non-iron ore economy of around 5.4 percent.

- Inflation is projected to rise in the medium term. Disruption of economic activity due to the Ebola outbreak and depreciation of the Leone are expected to push up inflation from 8.3 percent in 2014 to 9.9 percent in 2015. Inflation is expected to decline gradually and hover around 5.4 percent in the long term.

- The overall fiscal deficit is projected to worsen in the short term and then improve over the medium-to-long term, with the domestic primary deficit increasing from around 2.3 percent of GDP in 2014 to 3.7 percent in 2015. Beginning in 2016, it is projected to improve to 2 percent in 2018, as a result of revenue measures and efforts to strengthen tax administration. It is projected to stabilize around 0.1 percent during 2021–35.

- The current account deficit is forecast to decline from 19.2 percent of GDP in 2014 to 13.2 percent in 2015 and then narrow over the medium-to-long term, consistent with the projected production profile of iron ore and other exports. The projected import dynamics broadly reflect the expected overall real GDP growth.

- FDI is projected to decline from 8 percent of GDP in 2014 to around 7 percent in 2015, growing in line with machinery imports. Despite the present difficulties in the mining sector, there are signs of renewed interest from foreign investors in non-mining sectors of the economy, particularly, in agriculture and manufacturing.

- External debt is projected to increase from 25.3 percent of GDP in 2014 to 32.0 percent in 2015, reflecting the augmentation of access to IMF financing. External debt will stabilize around 20 percent of GDP in the long run.

- Domestic debt is projected to rise from 10.6 percent of GDP in 2014, to around 19 percent by the end of the projection period, mainly reflecting increased domestic borrowing to finance public investment.

5. **Compared with the previous DSA, the key changes in this updated DSA are (Text Table 1)**: (i) real GDP growth is expected to be much lower in the short term, due to the effects

of the Ebola epidemic and the suspension of iron-ore production. The economic contraction in 2015 is projected to accelerate from -12.8 percent to -21.5 percent, and in 2016, the growth will be zero instead of 8.4 percent increase (ii) a higher primary deficit, particularly in the short run (while lower in the long run), driven by Ebola and iron-ore-related revenue shortfalls and expanded spending needs; (iii) a higher level of domestic borrowing, in line with a weaker fiscal position; and (iv) the external current account deficit is largely unchanged in 2015 compared to the previous DSA, but is slightly worse in 2016, due mostly to the loss of minerals exports.

Text Table 1. Selected Economic Indicators, 2014–35
(Percent of GDP, unless otherwise indicated)[1]

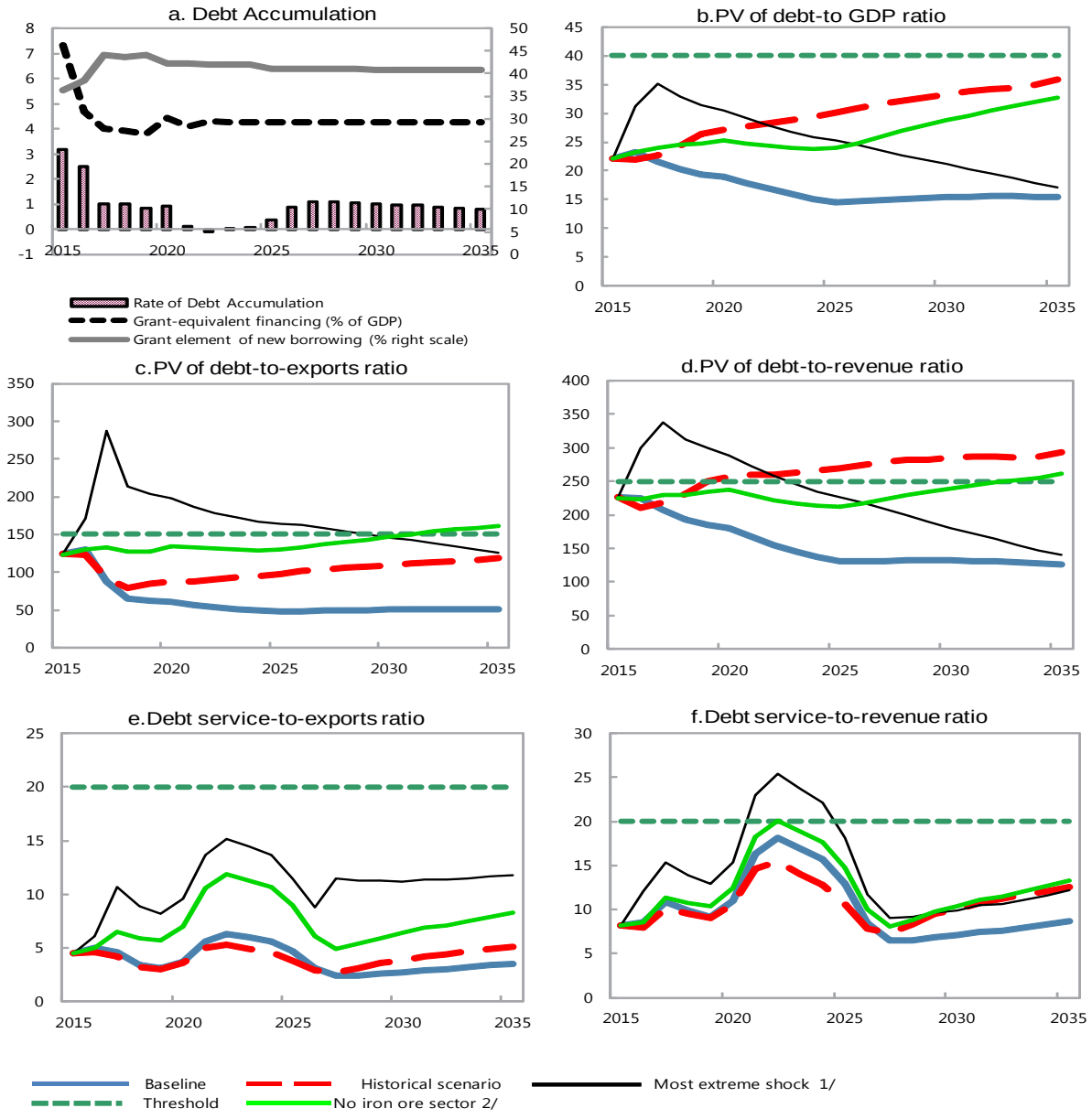
	2014	2015	2016	2017	2018	2019	2020	Long Term[2]
Real GDP Growth (in percent)								
Current DSA	4.6	-21.5	0.1	19.6	17.5	6.2	6.3	5.4
Previous DSA	6.0	-12.8	8.4	8.9	6.0	6.0	6.0	5.4
Primary fiscal deficit								
Current DSA	2.3	3.7	4.0	2.0	1.9	1.3	1.2	0.1
Previous DSA	2.0	3.5	4.5	3.7	3.3	3.2	3.0	1.8
Current account deficit								
Current DSA	19.2	13.2	11.9	10.5	7.7	5.8	8.3	4.4
Previous DSA	7.4	13.0	8.3	7.1	6.8	6.1	5.8	4.3
Foreign direct investment								
Current DSA	8.0	7.1	6.7	8.4	7.4	6.6	6.5	3.5
Previous DSA	4.6	3.0	5.6	4.5	4.0	3.3	3.0	3.2
External debt								
Current DSA	25.3	32.0	33.1	30.2	27.9	26.4	25.6	21.1
Previous DSA	28.2	33.7	30.5	27.7	26.5	25.9	25.4	19.9
Domestic debt								
Current DSA	10.6	12.5	15.3	15.0	14.7	15.5	16.2	18.8
Previous DSA	10.7	12.6	12.5	12.6	13.0	13.6	14.0	19.5

Sources: The Sierra Leone Authorities and IMF staff projections.

1/ GDP includes iron ore activity.

2/ For the current DSA, the long term covers the period 2021-35, and for the previous DSA it covers the period 2020-34.

Figure 1. Sierra Leone: Indicators of Public and Publicly Guaranteed External Debt under Alternative Scenarios, 2015–35[1]

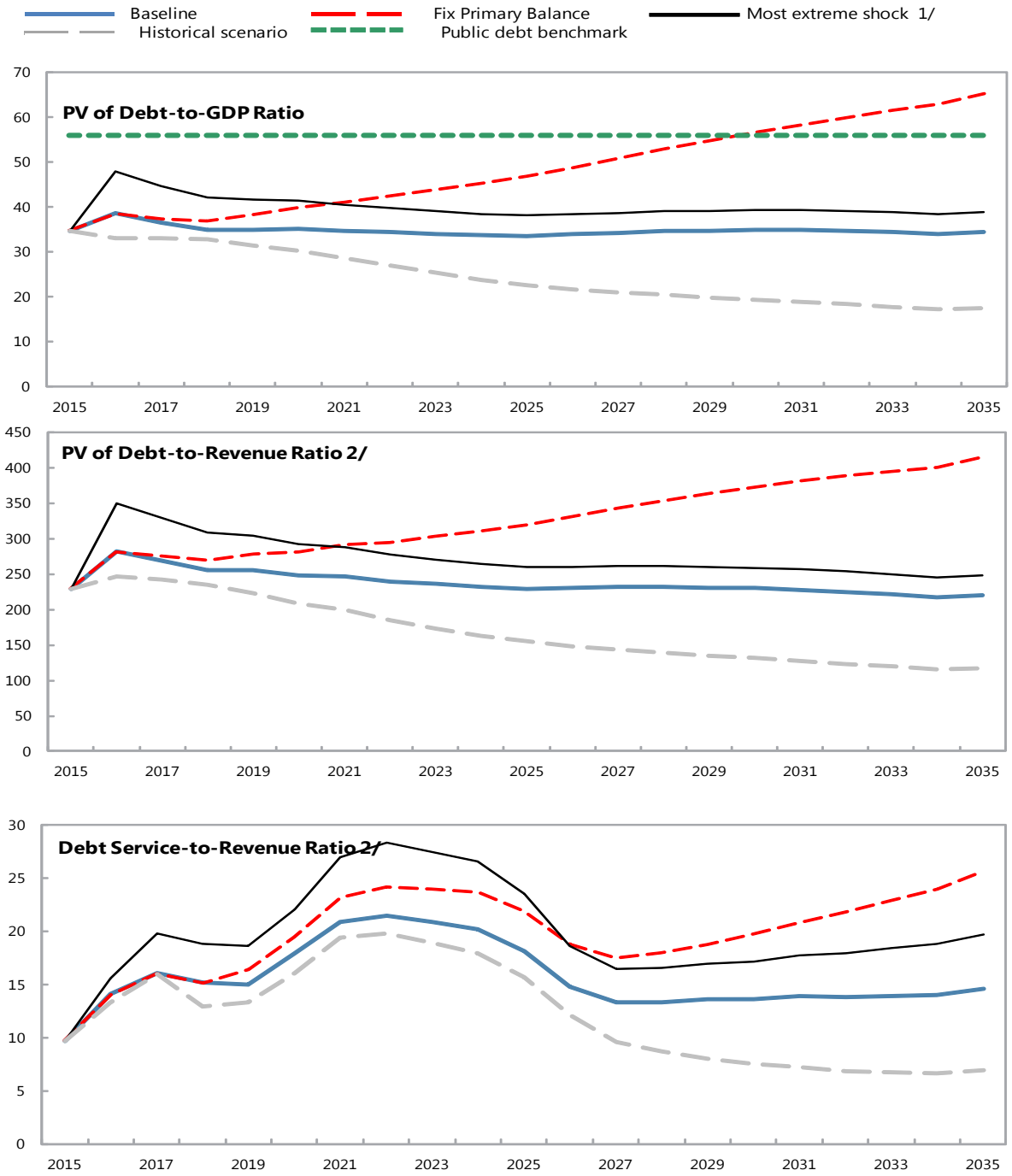


Sources: Country authorities; and staff estimates and projections.

1/ The most extreme stress test is the test that yields the highest ratio on or before 2025. In figure b. it corresponds to a Non-debt flows shock; in c. to a Exports shock; in d. to a Non-debt flows shock; in e. to a Exports shock and in figure f. to a One-time depreciation shock

2/ Iron ore production is permanently shut down

Figure 2. Sierra Leone: Indicators of Public Debt under Alternative Scenarios, 2015–35[1]



Sources: Country authorities; and staff estimates and projections.
1/ The most extreme stress test is the test that yields the highest ratio on or before 2025.
2/ Revenues are defined inclusive of grants.

Table 1. Sierra Leone: External Debt Sustainability Framework, Baseline Scenario, 2012–35[1]

	Actual			Historical Average 6/	Standard Deviation 6/	Projections									
	2012	2013	2014			2015	2016	2017	2018	2019	2020	2015-2020 Average	2025	2035	2021-2035 Average
External debt (nominal) 1/	**26.2**	**21.5**	**25.3**			**32.0**	**33.1**	**30.2**	**27.9**	**26.4**	**25.6**		**22.5**	**22.1**	
of which: public and publicly guaranteed (PPG)	26.2	21.5	25.3			32.0	33.1	30.2	27.9	26.4	25.6		22.5	22.1	
Change in external debt	-7.1	-4.7	3.8			6.7	1.1	-2.9	-2.3	-1.5	-0.7		-0.2	-0.2	
Identified net debt-creating flows	7.7	4.4	11.7			12.3	5.4	-3.6	-4.3	-2.2	0.4		0.4	-1.6	
Non-interest current account deficit	**31.7**	**17.6**	**19.2**	**15.8**	**13.8**	**13.3**	**11.9**	**10.5**	**7.7**	**5.8**	**8.3**		**4.4**	**2.3**	4.2
Deficit in balance of goods and services	33.3	10.4	28.8			25.4	20.6	13.2	5.8	5.2	5.0		4.9	5.0	
Exports	32.4	36.3	31.8			17.9	17.9	24.7	31.0	31.2	31.0		30.6	30.1	
Imports	65.8	46.6	60.6			43.3	38.5	38.0	36.7	36.4	36.1		35.5	35.1	
Net current transfers (negative = inflow)	-6.2	-4.2	-17.5	-7.3	4.0	-14.7	-10.2	-5.1	-5.0	-5.0	-5.0		-5.4	-6.9	-5.8
of which: official	-2.5	-1.0	-14.8			-11.5	-7.3	-2.2	-2.2	-2.3	-2.4		-2.9	-4.6	
Other current account flows (negative = net inflow)	4.6	11.4	7.9			2.6	1.6	2.3	7.0	5.6	8.3		4.9	4.2	
Net FDI (negative = inflow)	**-16.7**	**-7.4**	**-8.0**	**-10.1**	**10.5**	**-7.1**	**-6.7**	**-8.4**	**-7.4**	**-6.6**	**-6.5**		**-3.5**	**-3.0**	-3.5
Endogenous debt dynamics 2/	**-7.4**	**-5.8**	**0.6**			**6.1**	**0.2**	**-5.6**	**-4.6**	**-1.4**	**-1.4**		**-0.4**	**-1.0**	
Contribution from nominal interest rate	0.2	0.0	0.0			0.2	0.2	0.1	0.1	0.1	0.2		0.7	0.2	
Contribution from real GDP growth	-3.9	-4.2	-1.0			6.0	0.0	-5.7	-4.7	-1.6	-1.5		-1.1	-1.2	
Contribution from price and exchange rate changes	-3.6	-1.6	1.6			...	...	...	...	...	...		...	...	
Residual (3-4) 3/	**-14.8**	**-9.1**	**-7.9**			**-2.1**	**-2.9**	**-1.3**	**0.5**	**1.3**	**-0.8**		**-0.4**	**1.5**	
of which: exceptional financing	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
PV of external debt 4/	...	...	17.3			22.2	23.3	21.6	20.2	19.4	19.0		14.6	15.4	
In percent of exports	...	...	54.4			124.0	130.5	87.4	65.3	62.3	61.3		47.8	51.2	
PV of PPG external debt	**...**	**...**	**17.3**			**22.2**	**23.3**	**21.6**	**20.2**	**19.4**	**19.0**		**14.6**	**15.4**	
In percent of exports	**...**	**...**	**54.4**			**124.0**	**130.5**	**87.4**	**65.3**	**62.3**	**61.3**		**47.8**	**51.2**	
In percent of government revenues	**...**	**...**	**171.7**			**226.4**	**224.1**	**208.1**	**192.3**	**184.3**	**179.8**		**131.3**	**126.2**	
Debt service-to-exports ratio (in percent)	**1.8**	**0.0**	**0.0**			**4.5**	**5.0**	**4.6**	**3.4**	**3.1**	**3.7**		**4.7**	**3.5**	
PPG debt service-to-exports ratio (in percent)	**1.8**	**0.0**	**0.0**			**4.5**	**5.0**	**4.6**	**3.4**	**3.1**	**3.7**		**4.7**	**3.5**	
PPG debt service-to-revenue ratio (in percent)	**5.0**	**0.0**	**0.0**			**8.2**	**8.6**	**10.9**	**9.9**	**9.2**	**11.0**		**12.9**	**8.7**	
Total gross financing need (Billions of U.S. dollars)	0.6	0.5	0.5			0.3	0.3	0.2	0.1	0.0	0.2		0.2	0.1	
Non-interest current account deficit that stabilizes debt ratio	38.8	22.4	15.4			6.6	10.8	13.4	10.0	7.3	9.1		4.6	2.5	
Key macroeconomic assumptions															
Real GDP growth (in percent)	15.2	20.7	4.6	7.7	5.7	-21.5	0.1	19.6	17.5	6.2	6.3	4.7	5.1	5.7	5.4
GDP deflator in US dollar terms (change in percent)	12.2	6.5	-6.8	4.9	6.7	16.0	4.3	-5.6	-4.6	2.4	1.5	2.3	0.9	0.3	0.6
Effective interest rate (percent) 5/	0.7	0.0	0.0	0.6	0.5	0.6	0.6	0.4	0.5	0.6	0.6	0.6	3.1	1.2	1.7
Growth of exports of G&S (US dollar terms, in percent)	125.2	43.7	-14.5	24.3	39.6	-48.8	4.3	56.0	40.4	9.4	7.5	11.5	5.4	6.0	5.9
Growth of imports of G&S (US dollar terms, in percent)	20.1	-8.9	26.7	26.8	35.7	-34.9	-7.3	11.4	8.5	7.7	7.0	-1.3	5.8	2.8	5.9
Grant element of new public sector borrowing (in percent)	...	...	...	...	...	36.3	38.4	44.0	43.7	44.1	42.2	41.5	41.1	40.7	41.2
Government revenues (excluding grants, in percent of GDP)	11.4	10.8	10.1			9.8	10.4	10.4	10.5	10.5	10.6		11.1	12.2	11.5
Aid flows (in Billions of US dollars) 7/	0.3	0.2	0.3			0.3	0.2	0.2	0.3	0.3	0.3		0.5	0.8	
of which: Grants	0.1	0.1	0.2			0.2	0.1	0.2	0.2	0.2	0.2		0.3	0.6	
of which: Concessional loans	0.1	0.1	0.1			0.1	0.1	0.1	0.1	0.1	0.1		0.1	0.2	
Grant-equivalent financing (in percent of GDP) 8/	...	...	...			7.3	4.7	4.0	3.9	3.8	4.4		4.3	4.3	4.3
Grant-equivalent financing (in percent of external financing) 8/	...	...	...			68.0	67.2	79.4	79.2	81.8	79.5		79.4	79.3	79.5
Memorandum items:															
Nominal GDP (Billions of US dollars)	3.8	4.9	4.7			4.3	4.5	5.1	5.7	6.2	6.7		9.0	16.3	
Nominal dollar GDP growth	29.2	28.5	-2.5			-8.9	4.4	12.9	12.1	8.8	7.9	6.2	6.1	6.1	6.1
PV of PPG external debt (in Billions of US dollars)			0.8			0.9	1.0	1.1	1.1	1.2	1.2		1.3	2.4	
(PVt-PVt-1)/GDPt-1 (in percent)						3.2	2.5	1.0	1.0	0.9	0.9	1.6	0.4	0.8	0.7
Gross workers' remittances (Billions of US dollars)	0.0	0.0	0.0			0.0	0.0	0.1	0.1	0.1	0.1		0.1	0.1	
PV of PPG external debt (in percent of GDP + remittances)	...	...	17.1			21.9	23.1	21.4	20.1	19.2	18.9		14.5	15.3	
PV of PPG external debt (in percent of exports + remittances)	...	...	52.7			116.7	122.9	84.1	63.5	60.7	59.7		46.8	50.5	
Debt service of PPG external debt (in percent of exports + remittances)	...	...	0.0			4.2	4.7	4.4	3.3	3.0	3.6		4.6	3.5	

Sources: Country authorities; and staff estimates and projections.

1/ Includes both public and private sector external debt.
2/ Derived as [r - g - ρ(1+g)]/(1+g+ρ+gρ) times previous period debt ratio, with r = nominal interest rate; g = real GDP growth rate, and ρ = growth rate of GDP deflator in U.S. dollar terms.
3/ Includes exceptional financing (i.e., changes in arrears and debt relief); changes in gross foreign assets; and valuation adjustments. For projections also includes contribution from price and exchange rate changes.
4/ Assumes that PV of private sector debt is equivalent to its face value.
5/ Current-year interest payments divided by previous period debt stock.
6/ Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.
7/ Defined as grants, concessional loans, and debt relief.
8/ Grant-equivalent financing includes grants provided directly to the government and through new borrowing (difference between the face value and the PV of new debt).

Table 2. Sierra Leone: Sensitivity Analysis for Key Indicators of Public and Publicly Guaranteed External Debt, 2015–35[1]

(Percent)

	Projections							
	2015	2016	2017	2018	2019	2020	**2025**	2035
PV of debt-to GDP ratio								
Baseline	22	23	22	20	19	19	**15**	15
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2015-2035 1/	22	22	23	24	26	27	**30**	36
A2. New public sector loans on less favorable terms in 2015-2035 2	22	24	22	21	21	21	**19**	22
A3. Alternative Scenario :[Customize, enter title]	22	23	24	25	25	25	**24**	33
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2016-2017	22	22	24	23	22	21	**16**	17
B2. Export value growth at historical average minus one standard deviation in 2016-2017 3/	22	25	31	29	28	27	**22**	17
B3. US dollar GDP deflator at historical average minus one standard deviation in 2016-2017	22	24	22	20	19	19	**14**	15
B4. Net non-debt creating flows at historical average minus one standard deviation in 2016-2017 4/	22	31	35	33	31	31	**25**	17
B5. Combination of B1-B4 using one-half standard deviation shocks	22	26	33	31	30	29	**23**	17
B6. One-time 30 percent nominal depreciation relative to the baseline in 2016 5/	22	32	30	28	27	26	**20**	21
PV of debt-to-exports ratio								
Baseline	124	130	87	65	62	61	**48**	51
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2015-2035 1/	124	123	92	79	85	88	**98**	119
A2. New public sector loans on less favorable terms in 2015-2035 2	124	132	90	69	67	67	**61**	74
A3. Alternative Scenario :[Customize, enter title]	124	130	133	127	128	135	**130**	161
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2016-2017	124	128	86	64	61	60	**46**	50
B2. Export value growth at historical average minus one standard deviation in 2016-2017 3/	124	171	287	214	204	199	**164**	126
B3. US dollar GDP deflator at historical average minus one standard deviation in 2016-2017	124	128	86	64	61	60	**46**	50
B4. Net non-debt creating flows at historical average minus one standard deviation in 2016-2017 4/	124	174	142	106	101	98	**82**	57
B5. Combination of B1-B4 using one-half standard deviation shocks	124	149	192	143	136	133	**110**	83
B6. One-time 30 percent nominal depreciation relative to the baseline in 2016 5/	124	128	86	64	61	60	**46**	50
PV of debt-to-revenue ratio								
Baseline	226	224	208	192	184	180	**131**	126
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2015-2035 1/	226	211	218	231	251	257	**270**	293
A2. New public sector loans on less favorable terms in 2015-2035 2	226	227	215	202	197	195	**166**	184
A3. Alternative Scenario :[Customize, enter title]	226	223	229	229	235	238	**212**	261
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2016-2017	226	216	235	217	208	201	**146**	141
B2. Export value growth at historical average minus one standard deviation in 2016-2017 3/	226	239	301	278	266	257	**199**	137
B3. US dollar GDP deflator at historical average minus one standard deviation in 2016-2017	226	233	208	193	184	179	**130**	125
B4. Net non-debt creating flows at historical average minus one standard deviation in 2016-2017 4/	226	300	338	312	299	289	**226**	141
B5. Combination of B1-B4 using one-half standard deviation shocks	226	252	318	293	281	272	**211**	142
B6. One-time 30 percent nominal depreciation relative to the baseline in 2016 5/	226	309	287	265	254	246	**179**	172

Table 2. Sierra Leone: Sensitivity Analysis for Key Indicators of Public and Publicly Guaranteed External Debt, 2015–35[1] (Concluded)

(Percent)

Debt service-to-exports ratio								
Baseline	5	5	5	3	3	4	**5**	4
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2015-2035 1/	5	5	4	3	3	4	**4**	5
A2. New public sector loans on less favorable terms in 2015-2035 2	5	5	5	4	3	4	**4**	5
A3. Alternative Scenario :[Customize, enter title]	5	5	7	6	6	7	**9**	8
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2016-2017	5	5	5	3	3	4	**5**	4
B2. Export value growth at historical average minus one standard deviation in 2016-2017 3/	5	6	11	9	8	10	**12**	12
B3. US dollar GDP deflator at historical average minus one standard deviation in 2016-2017	5	5	5	3	3	4	**5**	4
B4. Net non-debt creating flows at historical average minus one standard deviation in 2016-2017 4/	5	5	5	4	4	4	**5**	6
B5. Combination of B1-B4 using one-half standard deviation shocks	5	5	7	6	5	6	**8**	8
B6. One-time 30 percent nominal depreciation relative to the baseline in 2016 5/	5	5	5	3	3	4	**5**	4
Debt service-to-revenue ratio								
Baseline	8	9	11	10	9	11	**13**	9
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2015-2035 1/	8	8	10	9	9	11	**11**	13
A2. New public sector loans on less favorable terms in 2015-2035 2	8	9	11	10	10	12	**12**	11
A3. Alternative Scenario :[Customize, enter title]	8	9	11	11	10	12	**15**	13
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2016-2017	8	8	13	11	11	13	**15**	10
B2. Export value growth at historical average minus one standard deviation in 2016-2017 3/	8	9	11	12	11	12	**14**	13
B3. US dollar GDP deflator at historical average minus one standard deviation in 2016-2017	8	9	11	10	9	11	**13**	9
B4. Net non-debt creating flows at historical average minus one standard deviation in 2016-2017 4/	8	9	12	12	11	13	**14**	14
B5. Combination of B1-B4 using one-half standard deviation shocks	8	8	12	12	11	13	**15**	14
B6. One-time 30 percent nominal depreciation relative to the baseline in 2016 5/	8	12	15	14	13	15	**18**	12
Memorandum item:								
Grant element assumed on residual financing (i.e., financing required above baseline) 6/	40	40	40	40	40	40	**40**	40

Sources: Country authorities; and staff estimates and projections.

1/ Variables include real GDP growth, growth of GDP deflator (in U.S. dollar terms), non-interest current account in percent of GDP, and non-debt creating flows.

2/ Assumes that the interest rate on new borrowing is by 2 percentage points higher than in the baseline., while grace and maturity periods are the same as in the baseline.

3/ Exports values are assumed to remain permanently at the lower level, but the current account as a share of GDP is assumed to return to its baseline level after the shock (implicitly assu an offsetting adjustment in import levels).

4/ Includes official and private transfers and FDI.

5/ Depreciation is defined as percentage decline in dollar/local currency rate, such that it never exceeds 100 percent.

6/ Applies to all stress scenarios except for A2 (less favorable financing) in which the terms on all new financing are as specified in footnote 2.

Table 3. Sierra Leone: Public Sector Debt Sustainability Framework, Baseline Scenario, 2012–35

(Percent of GDP, unless otherwise indicated)

	Actual					Estimate	Projections								
	2012	2013	2014	Average 5/	Standard Deviation 5/	2015	2016	2017	2018	2019	2020	2015-20 Average	2025	2035	2021-35 Average
Public sector debt 1/	36.9	30.8	35.9			44.5	48.4	45.2	42.6	41.9	41.8		41.4	41.3	
of which: foreign-currency denominated	26.3	21.5	25.3			32.0	33.1	30.2	27.9	26.4	25.6		22.5	22.1	
Change in public sector debt	-8.0	-6.1	5.1			8.6	3.9	-3.2	-2.6	-0.7	-0.1		0.2	0.4	
Identified debt-creating flows	-6.0	-4.9	5.1			6.0	2.1	-2.9	-2.3	-1.4	-0.8		-1.2	-2.4	
Primary deficit	3.5	1.1	2.3	-0.9	7.6	3.7	4.0	2.0	1.9	1.3	1.2	2.3	0.3	-0.6	0.1
Revenue and grants	15.2	13.4	14.4			15.1	13.7	13.6	13.6	13.7	14.2		14.6	15.7	
of which: grants	3.8	2.6	4.3			5.3	3.2	3.2	3.1	3.1	3.6		3.5	3.5	
Primary (noninterest) expenditure	18.7	14.5	16.7			18.8	17.7	15.6	15.6	14.9	15.3		14.9	15.1	
Automatic debt dynamics	-8.7	-5.9	3.0			2.5	-1.9	-4.8	-4.2	-2.7	-2.0		-1.4	-1.8	
Contribution from interest rate/growth differential	-5.8	-5.0	-0.2			7.4	-0.5	-7.0	-6.0	-2.7	-2.3		-1.2	-1.7	
of which: contribution from average real interest rate	0.1	1.3	1.1			-2.4	-0.4	0.9	0.7	-0.2	0.2		0.8	0.5	
of which: contribution from real GDP growth	-5.9	-6.3	-1.4			9.8	0.0	-7.9	-6.7	-2.5	-2.5		-2.0	-2.2	
Contribution from real exchange rate depreciation	-2.9	-0.9	3.2			-4.8	-1.4	2.2	1.8	0.0	0.3		...	...	
Other identified debt-creating flows	-0.8	-0.1	-0.2			-0.2	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Privatization receipts (negative)	-0.1	-0.1	-0.1			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Recognition of implicit or contingent liabilities	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Debt relief (HIPC and other)	-0.8	0.0	-0.1			-0.2	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Other (specify, e.g. bank recapitalization)	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Residual, including asset changes	-2.0	-1.2	0.1			2.6	1.7	-0.3	-0.3	0.7	0.8		1.3	2.9	
Other Sustainability Indicators															
PV of public sector debt	...	...	27.9			34.7	38.6	36.6	34.9	34.9	35.2		33.6	34.6	
of which: foreign-currency denominated	...	...	17.3			22.2	23.3	21.6	20.2	19.4	19.0		14.6	15.4	
of which: external	...	...	17.3			22.2	23.3	21.6	20.2	19.4	19.0		14.6	15.4	
PV of contingent liabilities (not included in public sector debt)	...	...	...			...	...	...	...	...	...		...	...	
Gross financing need 2/	5.6	2.4	3.1			5.2	5.9	4.1	4.0	3.3	3.7		2.9	1.6	
PV of public sector debt-to-revenue and grants ratio (in percent)	...	...	193.7			229.2	282.6	269.2	256.1	255.5	248.4		229.7	220.1	
PV of public sector debt-to-revenue ratio (in percent)	...	...	277.1			354.4	370.7	352.0	332.1	331.7	333.1		301.8	283.2	
of which: external 3/	...	...	171.7			226.4	224.1	208.1	192.3	184.3	179.8		131.3	126.2	
Debt service-to-revenue and grants ratio (in percent) 4/	13.9	9.4	5.7			9.7	14.1	16.0	15.2	15.0	17.9		18.1	14.6	
Debt service-to-revenue ratio (in percent) 4/	18.6	11.7	8.2			15.0	18.5	21.0	19.7	19.5	24.0		23.8	18.7	
Primary deficit that stabilizes the debt-to-GDP ratio	11.5	7.2	-2.9			-4.9	0.1	5.2	4.5	2.0	1.2		0.1	-1.1	
Key macroeconomic and fiscal assumptions															
Real GDP growth (in percent)	15.2	20.7	4.6	7.7	5.7	-21.5	0.1	19.6	17.5	6.2	6.3	4.7	5.1	5.7	5.4
Average nominal interest rate on forex debt (in percent)	0.7	0.0	0.0	0.8	0.6	0.6	0.6	0.4	0.5	0.6	0.6	0.6	3.1	1.2	1.7
Average real interest rate on domestic debt (in percent)	4.8	8.2	9.5	0.6	5.4	-17.1	-1.1	10.2	9.0	1.8	3.5	1.0	0.8	1.5	1.1
Real exchange rate depreciation (in percent, + indicates depreciation)	-10.0	-4.2	15.6	-2.1	10.3	-15.1	...	...	...	...	...	...	...	...	...
Inflation rate (GDP deflator, in percent)	12.0	6.5	-0.2	10.6	5.6	28.3	10.4	-1.9	-0.9	6.4	6.3	8.1	6.5	5.9	6.2
Growth of real primary spending (deflated by GDP deflator, in percent)	9.8	-6.4	20.4	2.5	7.4	-11.3	-6.2	5.3	17.8	1.8	9.1	2.7	5.1	5.6	5.3
Grant element of new external borrowing (in percent)	...	...	...	...	...	36.3	38.4	44.0	43.7	44.1	42.2	41.5	41.1	40.7	...

Sources: Country authorities; and staff estimates and projections.

1/ [Indicate coverage of public sector, e.g., general government or nonfinancial public sector. Also whether net or gross debt is used.]

2/ Gross financing need is defined as the primary deficit plus debt service plus the stock of short-term debt at the end of the last period.

3/ Revenues excluding grants.

4/ Debt service is defined as the sum of interest and amortization of medium and long-term debt.

5/ Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.

Table 4. Sierra Leone: Sensitivity Analysis for Key Indicators of Public Debt, 2015–35

	Projections							
	2015	2016	2017	2018	2019	2020	2025	2035
PV of Debt-to-GDP Ratio								
Baseline	35	39	37	35	35	35	34	35
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	35	33	33	33	31	30	23	18
A2. Primary balance is unchanged from 2015	35	38	37	37	38	40	47	65
A3. Permanently lower GDP growth 1/	35	39	38	37	37	38	42	61
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2016-2017	35	38	43	42	43	44	46	55
B2. Primary balance is at historical average minus one standard deviations in 2016-2017	35	40	41	39	39	39	37	37
B3. Combination of B1-B2 using one half standard deviation shocks	35	36	40	38	39	40	41	46
B4. One-time 30 percent real depreciation in 2016	35	48	45	42	42	41	38	39
B5. 10 percent of GDP increase in other debt-creating flows in 2016	35	45	42	40	40	40	38	37
PV of Debt-to-Revenue Ratio 2/								
Baseline	229	283	269	256	255	248	230	220
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	229	246	242	234	224	208	155	117
A2. Primary balance is unchanged from 2015	229	281	276	270	279	281	320	415
A3. Permanently lower GDP growth 1/	229	286	276	266	270	266	278	366
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2016-2017	229	277	305	297	302	298	306	339
B2. Primary balance is at historical average minus one standard deviations in 2016-2017	229	295	302	285	282	274	252	233
B3. Combination of B1-B2 using one half standard deviation shocks	229	268	286	277	280	275	272	289
B4. One-time 30 percent real depreciation in 2016	229	350	329	309	304	292	260	248
B5. 10 percent of GDP increase in other debt-creating flows in 2016	229	327	311	292	290	281	259	236
Debt Service-to-Revenue Ratio 2/								
Baseline	10	14	16	15	15	18	18	15
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	10	13	16	13	13	16	16	7
A2. Primary balance is unchanged from 2015	10	14	16	15	16	20	22	26
A3. Permanently lower GDP growth 1/	10	14	16	16	16	19	21	22
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2016-2017	10	14	18	17	18	21	22	22
B2. Primary balance is at historical average minus one standard deviations in 2016-2017	10	14	16	17	18	19	19	16
B3. Combination of B1-B2 using one half standard deviation shocks	10	14	17	16	17	20	20	19
B4. One-time 30 percent real depreciation in 2016	10	16	20	19	19	22	24	20
B5. 10 percent of GDP increase in other debt-creating flows in 2016	10	14	17	21	16	19	19	17

Sources: Country authorities; and staff estimates and projections.

1/ Assumes that real GDP growth is at baseline minus one standard deviation divided by the square root of the length of the projection period.

2/ Revenues are defined inclusive of grants.